STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 14: Waves

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Transverse oscillation relative to wave travel? [1]
2. Sound in air type? [1]
3. Wave speed relation? [1]
4. Wave number k? [1]
5. Superposition principle? [1]
6. Reflection at fixed end? [1]
7. Beat frequency? [1]
8. What is amplitude? [1]

Section B 2-mark questions - 6 x 2 = 12 marks

9. f=50Hz, lambda=2m: speed? [2]
10. v=340m/s, f=170Hz: wavelength? [2]
11. 256Hz and 260Hz beats? [2]
12. Wavelength halves at fixed speed: frequency? [2]
13. Reflection at free end: inversion? [2]
14. Wave y=A sin(kx-omegat) travels which way? [2]

Section C 3-mark questions - 4 x 3 = 12 marks

15. v=24m/s, lambda=.8m: f and period? [3]
16. Displacements +3 cm and -1 cm overlap: resultant? [3]
17. 300Hz and 304Hz: beat frequency and period? [3]
18. How do waves transfer energy without permanent medium transport? [3]

Section D 4-mark questions - 2 x 4 = 8 marks

19. Wave travels 12m in .4s with f=5Hz: speed/wavelength? [4]
20. Explain beats for 440Hz and 442Hz. [4]

End of question paper.

FORMULA AND EQUATION REFERENCE

Chapter-specific equations and relations

- Wave speed $v=f\lambda$; period $T=1/f$; angular frequency $\omega=2\pi f$; wave number $k=2\pi/\lambda$.
- Progressive wave travelling in +x: $y=A\sin(kx-\omega t)$; travelling in -x: $y=A\sin(kx+\omega t)$.
- For a string with tension F and linear density μ, wave speed $v=\sqrt{F/\mu}$.
- Beat frequency $f_b=|f_1-f_2|$; beat interval $T_b=1/f_b$.
- Superposition: resultant displacement is the algebraic sum of individual displacements.
- For a fixed end, reflected pulse is inverted; for an ideal free end, it is not inverted.
- Wave energy travels through the medium while particles oscillate about equilibrium; they do not undergo permanent net transport.

Use equations only when their assumptions apply; keep units consistent.

STEP-BY-STEP SOLUTIONS

Concept questions are answered directly; numerical problems show the governing equation and substitution.

Question 1

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Perpendicular.
Final answer: Perpendicular.

Question 2

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Longitudinal.
Final answer: Longitudinal.

Question 3

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: v=flambda.
Final answer: v=flambda.

Question 4

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: 2pi/lambda.
Final answer: 2pi/lambda.

Question 5

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Resultant displacement is the algebraic sum.
Final answer: Resultant displacement is the algebraic sum.

Question 6

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Pulse is inverted.
Final answer: Pulse is inverted.

Question 7

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: |f_1-f_2|.
Final answer: |f_1-f_2|.

Question 8

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Maximum displacement from equilibrium.
Final answer: Maximum displacement from equilibrium.

Question 9

Working: v=f lambda=50(2)=100 m/s.

Final answer: 100 m/s.

Question 10

Working: lambda=v/f=340/170=2 m.

Final answer: 2 m.

Question 11

Working: Beat frequency=|f_1-f_2|=|260-256|=4 Hz.

Final answer: 4 Hz.

Question 12

Working: At fixed wave speed v=f lambda. If lambda is halved, frequency must double to keep v constant.

Final answer: Doubles.

Question 13

Working: At an ideal free end, the reflected pulse is not inverted; it returns with the same displacement sense.

Final answer: No.

Question 14

Working: The phase is kx-omega t. A point of constant phase satisfies x=(omega/k)t+constant, which moves toward increasing x, so the wave travels in +x.

Final answer: +x direction.

Question 15

Working: f=v/lambda=24/0.8=30 Hz. Period T=1/f=1/30=0.0333 s.

Final answer: 30Hz; 1/30s.

Question 16

Working: By superposition, resultant displacement=(+3 cm)+(-1 cm)=+2 cm.

Final answer: +2 cm.

Question 17

Working: Beat frequency=|304-300|=4 Hz. Time between beats T_b=1/f_b=1/4=0.25 s.

Final answer: 4 Hz; .25 s.

Question 18

Working: The medium’s particles oscillate locally about equilibrium. Their interactions pass the disturbance and energy along the medium, while particles do not travel permanently with the wave.

Final answer: Particles oscillate locally while the disturbance propagates.

Question 19

Working: Wave speed v=distance/time=12/0.4=30 m/s. Wavelength lambda=v/f=30/5=6 m.

Final answer: 30m/s; lambda=6m.

Question 20

Working: Beat frequency=|442-440|=2 Hz. Loudness rises and falls twice per second due to alternating constructive and destructive interference.

Final answer: Alternating loudness occurs at 2 beats per second.